

From Causal Auditing to Value Maximization for Core Distribution Assets — Part II: Switchgear Management

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Abstract—With the evolution of smart distribution grids, the predictive maintenance of heavy assets is shifting from calendar-based population management to data-driven individualized management. Focusing on the core primary asset in distribution grids—switchgear—this paper proposes a dynamic retirement and precise life-extension framework based on Remaining Useful Life (RUL) auditing. This study establishes an asymmetric resource scheduling boundary anchored on the “High-Voltage Isolation Cost,” rejecting the pseudo-proposition that all components require monitoring. Instead, it proposes a generalized auditing architecture targeting two core heavy assets: Air-Insulated Switchgear (AIS) and Gas-Insulated Switchgear (GIS). Asset owners can independently or flexibly deploy targeted audits for these equipments based on actual grid configuration needs. By introducing the signature “6-Mirror Panel Auditing” mechanism of the *Bianque System*, data undergoes blind-selection trials within survival analysis models. This process physically and reversely reveals the disparate degradation mechanisms: the “middle-age crisis” (an inverted-U hazard rate) of AIS mechanical operating mechanism fatigue, and the irreversible monotonic degradation of GIS chemical seal aging. Furthermore, this paper proposes a “10-Decile Sequential Soft-Threshold Calibration and Dual Retirement Boundary” mechanism. Combined with Optimal Stopping Theory, it achieves highly reliable interventions where the process relies on economic calculation, while the bottom line is secured by a mandatory scheduling alarm at the 9th decile. This avoids additional losses from unplanned outages caused by unexpected failures. Ultimately, the system translates survival curves into precise commercial decision points, maximizing the Return on Investment (ROI) across the lifecycle of heavy distribution grid assets.

Index Terms—Predictive Maintenance, Survival Analysis, Remaining Useful Life (RUL), Optimal Stopping Theory, Switchgear.

I. INTRODUCTION

AS the world moves toward intelligence and high-penetration renewable energy integration [1], the smart distribution grid, acting as the nexus between the main power grid and end-users, faces unprecedented challenges in power supply reliability [2]. Within the massive fleet of equipment in distribution networks, core primary heavy assets like switchgear not only undertake crucial high-voltage clearance and isolation tasks, but their full-lifecycle maintenance and replacement also account for massive Capital Expenditure (CapEx) and Operational Expenditure (OpEx) for grid utilities [3]. However, due to historical investment cycles, a large number of heavy assets are collectively entering their aging

phase. In this context, Predictive Maintenance (PdM) based on sensor networks and survival models is rapidly replacing the traditional “scheduled maintenance” paradigm, becoming a shared focal point in both academia and industry [4], [5].

To address these challenges and overcome the limitations of traditional black-box AI prediction algorithms that lack physical interpretability, this project deeply inherits the underlying Clark Paradigm philosophy of “physical causal auditing and value maximization” established by the author in the offshore wind sector [6]. Consequently, the *Bianque System* has been developed for core distribution grid assets. The core soul of this system lies in “preventative healing and economic calculation”: for extremely short-cycle catastrophic failures caused by insulation breakdown or severe short-circuits, the system automatically isolates and hands them over to the homologous *Laplace Platform* [7] for asymmetric deterministic interception; whereas for long-cycle degradation failures with progressive physical trajectories, survival models are utilized for refined lifespan auditing. Together, they form a seamlessly interlocked dual-engine industrial diagnostic ecosystem.

The initial design intention of the *Bianque System* is to break through the passive response mode of “single rigid dead thresholds” currently prevalent in distribution grid asset management. Traditional operations and maintenance (O&M) schemes typically dictate that alarms are triggered by the backend only when the mechanical action of an AC circuit breaker severely times out (referencing the **IEC 62271-100** standard [8] and mechanical condition monitoring literature [9]) or when the SF6 gas pressure of a GIS drops below the minimum functional locking redline (referencing the **IEC 62271-203** standard [10] and leakage mechanism literature [11]). This paradigm is essentially “collecting the corpse” for the equipment, frequently forcing grids to face exceedingly expensive emergency unplanned outages and hefty fines. To break this deadlock, the system abandons the one-size-fits-all dead threshold approach. Instead, it utilizes remarkably inexpensive non-intrusive online data (e.g., micro-current waveforms from secondary control circuits) to acutely capture degradation slopes. It customizes an exclusive survival curve for every single heavy asset and feeds this into optimal stopping theory equations. This precisely outputs the optimal power outage and replacement time window, assisting owners in maximizing their ROI within safe boundaries.

As the second part of the *Bianque System* series research for distribution grids, this paper focuses on the switchgear family. It is crucial to define that although Solid-State Switchgear

(SIS) based on power electronic devices represents the future direction of flexible control, its penetration rate in the current massive legacy grids is still in its infancy. Moreover, its core failure mechanism (semiconductor thermal stress cycling fatigue) has a fundamental physical gap compared to the mechanical or chemical degradation of traditional heavy assets. Therefore, SIS is temporarily excluded from this evaluation scope. This paper will concentrate computational resources on conquering the two legacy mainstays currently facing the greatest retirement challenges in distribution grids: Air-Insulated Switchgear (AIS) and Gas-Insulated Switchgear (GIS). To reveal the true degradation laws of equipment under complex environments, the system deeply processes multi-source Right-Censored Data, which includes a large number of healthy surviving samples. Through a panel data blind-selection mechanism, the underlying physical aging laws are reversely restored at the mathematical level. On this basis, the probability curve of individual Remaining Useful Life (RUL) is substituted into the Optimal Stopping Theory equation [12]. This accurately deduces the equipment's maintenance time boundary and generates predictive reports for the owner's final confirmation of outage scheduling, thereby achieving the optimal commercial balance between asset residual value extraction and outage intervention costs.

In summary, the main contributions of this paper can be classified into three points:

- **Reverse derivation of physical truths covering two core assets:** Breaking the intuitive prejudice that “the older it gets, the easier it breaks,” this study uses data and statistical laws to confirm the distinct underlying physical degradation mechanisms: the “middle-age crisis” (inverted-U hazard rate) of AIS mechanical fatigue, and the chemical seal aging (monotonically increasing hazard rate) of GIS.
- **6-Mirror panel blind-selection validation under dual physical mechanisms:** Following the 6-mirror horse-racing mechanism established in PART I of this series [13], six classical and nonlinear models, including Cox, Weibull, Log-logistic, and Random Survival Forests, are introduced without presetting any prior distributions. This allows multi-source right-censored data to compete fairly within a unified framework, ensuring the objectivity of causal inference.
- **Dual retirement economic redline based on 10-Decile calibration:** Relying on the classic P-F interval concept [14] in Reliability-Centered Maintenance (RCM) theory, an innovative “Data + Time” dual-insurance P-point trigger mechanism is proposed. Furthermore, a 10-Decile sequential soft-threshold calibration method is designed. This method deeply integrates Bayesian error convergence with the partial differential equation solving of optimal stopping theory, achieving highly reliable interventions where “the process relies on economic calculation to extract residual value, while the bottom line relies on mandatory scheduling at the 9th decile,” ultimately providing operators with precise decision points for maximizing ROI.

II. RESEARCH BOUNDARY AND AUDITING ARCHITECTURE FOR TWO CORE ASSETS

A. System Inclusion Criteria: Auditing Boundary Based on Commercial ROI and Outage Isolation Costs

In smart distribution grids, from a broad reliability perspective, theoretically all equipment nodes possess potential value for online monitoring. However, at the industrial deployment level, whether to introduce high-dimensional sensor data streams and complex survival analysis models for specific equipment is essentially a commercial game based on Return on Investment (ROI) and intervention costs.

Internal equipment in distribution grids exhibits structural magnitude differences in “failure penalties” and “intervention costs.” For highly modularized, low-voltage secondary systems or power electronic devices (such as microcomputer protection relays, SVG plug-ins, etc.), their physical structures mostly support “plug-and-play” and often possess deep redundancy. The direct replacement cost after failure is extremely low, and in most cases, there is no need to switch or modify the main high-voltage lines. If expensive predictive computational power is blindly piled onto these low-intervention-cost nodes, the marginal monitoring cost generated will far exceed the asset's residual value, leading to a trap of negative commercial returns.

In stark contrast are core primary heavy assets such as AIS and GIS. The fundamental O&M pain point of such equipment is that any substantive maintenance, core component replacement, or complete decommissioning must strictly execute a **High-Voltage Clearance / Isolation** operation. This operation not only involves the extremely cumbersome construction of electrical safety boundaries but directly triggers large-scale power interruptions, exposing grid enterprises to exorbitant unplanned outage fines and negative operational assessments. Faced with such high-cost assets, utilizing survival analysis to predict degradation in advance and accurately lock onto the optimal outage time window yields a defensive moat value in mitigating economic losses that is absolutely decisive.

Therefore, constrained by both economics and engineering logic, the *Bianque System* does not fall into the pseudo-proposition of “monitoring everything,” but prudently defines an auditing boundary based on “outage isolation costs.” The system strategically focuses the computational power of high-order survival models entirely on core primary heavy assets that mandate high-voltage outage isolation. For low-dimensional modular equipment, it recommends retreating to more commercially optimal “Run-to-Failure” or basic dead-threshold alarm strategies. This asymmetric resource scheduling mechanism fundamentally guarantees the highest financial return when the predictive maintenance paradigm is scaled up to the grid level.

B. Flexible Deployment Architecture and Deconstruction of Underlying Physical Truths

The framework constructed in this study grants asset owners tremendous autonomy. Owners can independently deploy monitoring for AIS or GIS, or any combination thereof, based on actual requirements. Adhering to the “Weakest Link Effect”

principle [15], the system respectively extracts the core physical fingerprints that determine the scrapping and replacement of the two asset classes. To mathematically deconstruct these two distinct physical truths, this paper employs the Hazard Function $h(t)$ from the standard survival analysis framework [16]. The function $h(t)$ describes the instantaneous probability density of equipment failing in the next instant, given that it has survived to time t , with a range of $[0, \infty)$. By statistically reconstructing the underlying variables, we can accurately distinguish between mechanical fatigue and chemical aging.

1) *AIS Auditing Standards and the Log-Logistic Distribution of Mechanical Fatigue:* Although AIS is exposed to complex operating environments, most of its failures and ultimate retirements stem from the jamming of the heavy spring mechanical operating mechanism [17]. This system selects the **closing/opening coil current delay and spring energy storage time** as its core observed covariates (the relevant mechanical operational boundaries follow the **IEC 62271-100** standard [8]). In numerical simulation deductions, it was discovered that the physical fatigue of the spring mechanism does not rise linearly but exhibits a typical “middle-age crisis.” To mathematically depict this non-monotonic characteristic, the system adopts a Log-logistic hazard function featuring an “inverted-U” shape:

$$h_{AIS}(t) = \frac{(\beta/\alpha)(t/\alpha)^{\beta-1}}{1 + (t/\alpha)^\beta} \quad (1)$$

Equation (1) originates from an Accelerated Failure Time (AFT) model variant in statistics [4], [18]. Here, t is the equipment’s operating time; $\alpha > 0$ is the Scale Parameter, whose physical meaning is the median life of the population; $\beta > 0$ is the Shape Parameter, which determines the severity of degradation.

To characterize the “middle-age crisis” of AIS, the shape parameter must satisfy $\beta > 1$. At this point, Equation (1) exhibits a delicate “numerator-denominator game.” The numerator contains $t^{\beta-1}$, representing the normal physical wear and jamming tendency accumulated by the spring mechanism over time; while the denominator contains t^β , whose temporal growth power is strictly greater than that of the numerator.

In the early stages of operation (small t), the denominator term $(t/\alpha)^\beta$ is extremely small and poses almost no damping effect. At this time, the mechanical wear in the numerator dominates, causing the instantaneous hazard rate $h_{AIS}(t)$ to climb rapidly, bringing the equipment into the high-incidence phase of the “middle-age crisis.” However, as operating time t continues to advance, the high-order exponential growth of t^β in the denominator begins to demonstrate overwhelming convergent dominance. In reliability physics, this corresponds to the “Survivor Effect”: if an AIS successfully endures the harshest mechanical run-in period without jamming, it proves that the individual’s assembly craftsmanship or physical operating conditions are significantly superior to the population average. For these robust surviving equipments, the strong suppression by the high-order denominator rapidly dilutes the instantaneous probability of sudden failure in the next moment, ultimately causing the hazard rate curve to show a gentle, long-tail decline after crossing the peak.

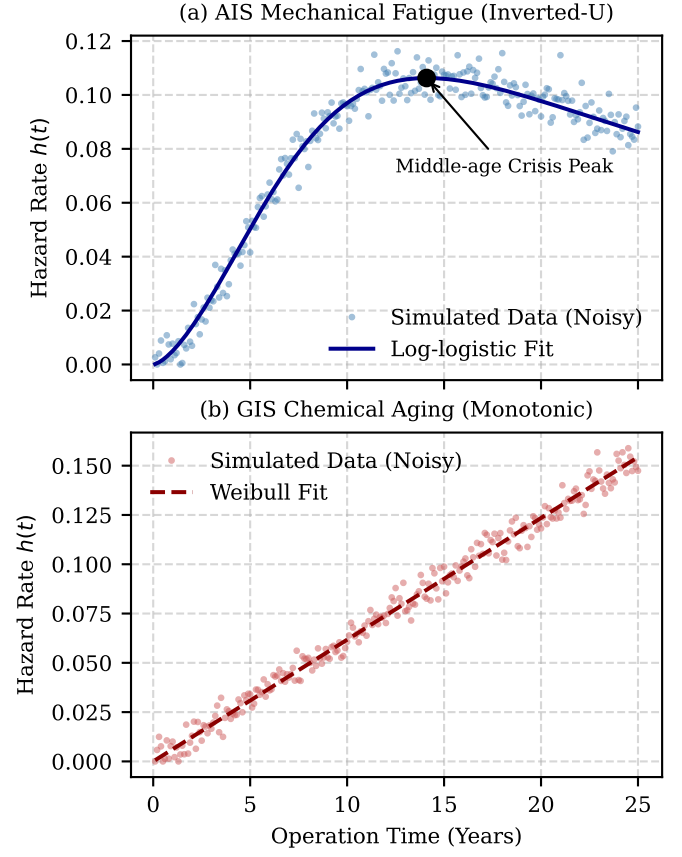


Fig. 1. Numerical simulation comparison of core degradation mechanisms for two types of switchgear. (a) Inverted-U mechanical fatigue characteristics of AIS based on Log-logistic distribution; (b) Monotonically increasing chemical aging characteristics of GIS based on Weibull distribution. Data in this figure originates from mathematical logic-based numerical simulation deductions, not real physical field measurements.

2) *GIS Auditing Standards and the Weibull Distribution of Chemical Seals:* Unlike AIS, GIS isolates external conditions through a fully enclosed architecture. Its weak link lies in SF6 gas micro-leakage caused by the aging of rubber sealing rings [19]. Therefore, the system selects the **SF6 pressure drop rate** as the core covariate (its minimum functional locking threshold follows the **IEC 62271-203** standard [10]).

It is worth noting that this system proactively discarded the equally concerning “moisture content” and “partial discharge (PD)” indicators during the feature engineering phase. Moisture was discarded because its sensor readings are highly susceptible to severe interference from external diurnal temperature differences, and the high environmental noise would directly destroy the monotonic convergence of the survival algorithm. PD was discarded due to its highly sporadic and sudden-breakdown characteristics, lacking a long-term predictable progressive degradation trajectory. Within the dual-track ecological architecture of this study, such transient, ultra-short-cycle insulation failures have been explicitly assigned to the *Laplace Platform* for interception. In contrast, the pressure drop rate is not only data-stable but also directly maps the absolute irreversible hardware lifespan baseline of sealing ring aging.

This aging of polymers is an irreversible process at the chemical level. To capture its monotonically increasing characteristic, the system adopts the classic Weibull hazard function [4]:

$$h_{GIS}(t) = \frac{\beta}{\alpha} \left(\frac{t}{\alpha} \right)^{\beta-1} \quad (2)$$

Equation (2) is similarly based on time t , scale parameter α , and shape parameter β . The greatest mathematical difference from Equation (1) is that the Weibull model completely discards the damping division structure. As long as the shape parameter is set to $\beta > 1$, its partial derivative with respect to time t is strictly positive due to the lack of high-order suppression from a t^β denominator. This pure multiplicative structure of the $t^{\beta-1}$ increasing term implies that the microscopic chemical aging of the sealing ring yields no “survivor effect” dividends. This structure mathematically and ruthlessly argues that the airtightness of high-voltage GIS is always in a state of irreversible, unidirectional deterioration.

3) *Numerical Simulation Validation of Degradation Mechanisms*: To intuitively demonstrate the heterogeneous attenuation paths guided by the aforementioned formulas, this paper generated simulated data conforming to these two distributions by overlaying Gaussian white noise based on the Monte Carlo method. Control validation was conducted under a unified time axis, and the results are shown in Fig. 1.

As shown in Fig. 1, the horizontal axis of both subplots represents equipment operating time (years), and the vertical axis represents the instantaneous hazard rate $h(t)$. The light-colored scatter points distributed at the bottom represent numerical sampling points generated after adding Gaussian noise ($\mathcal{N}(0, 0.005)$), used to simulate sensor acquisition fluctuations under complex physical conditions. The solid dark-blue line and dashed dark-red line respectively represent the main trend curves fitted and generated by the algorithms using Equation (1) and Equation (2).

In Fig. 1(a), it can be clearly observed that the solid dark-blue line exhibits a significant “Middle-age Crisis Peak” during the mid-term of operation (indicated by the black circle and arrow in the figure), and then gradually declines. In contrast, in Fig. 1(b), the dashed dark-red line presents a continuously upward parabolic form, directly reflecting the irreversible chemical deterioration process of the GIS sealing ring.

Results and Causal Inference Arguments: The aforementioned simulation results not only corroborate each other mathematically but also point out the stereotypical prejudice in traditional O&M that “equipment lifespan and failure risk have a simple linear positive correlation” from an engineering logic perspective. This result establishes the core argument of this study: for AIS assets, blind periodic full-scale replacement is a massive waste of funds. The optimal predictive maintenance strategy is to focus on escorting them smoothly through the early and mid-term mechanical run-in phase. Once the peak is crossed, the survivor effect can be utilized to drastically reduce intervention frequency and maximize the extraction of asset residual value. Conversely, for GIS assets, any fluke mentality of “outlasting the high-incidence period” is ex-

tremely dangerous. Its irreversible unidirectional deterioration requires the system to continuously and strictly tighten the economic redline for safety interventions as equipment age increases. This heterogeneous causal inference directly constructs the physical legitimacy of the “Dynamic Soft-Threshold Approaching Based on the P-F Interval” in the subsequent chapter.

III. CORE METHODOLOGY: 6-MIRROR PANEL AUDITING AND PHYSICAL DISCOVERY

The system deeply integrates the philosophy of “physical causal auditing” with the survival model classification framework, inheriting and deploying the “6-Mirror Model Panel Auditing” mechanism established in PART I of this series [13].

A. Data Integration and Censoring Handling: Overcoming Survivorship Bias

In complex actual operating environments, most switchgears do not experience physical failure by the end of a given observation period. Such data, which only records “safe operating time” but whose ultimate lifespan endpoint is unknown, is termed **Right-censored** data in statistics. Conventional machine learning algorithms, when dealing with such data lacking explicit “fault labels,” are usually forced to discard them directly. This processing method leads to severe statistical consequences—Survivorship Bias. Because the model can only learn based on early failure samples that have already occurred, completely discarding information from long-term healthy service equipment, the predicted life expectancy will be severely underestimated.

The core logic of introducing the survival analysis framework into this system is precisely to solve this bias problem from its mathematical roots. To handle right-censored data, the system employs Maximum Likelihood Estimation (MLE [20]) with event indicators as the basic objective function. Its log-likelihood meta-equation is deconstructed as follows:

$$\ln L(\theta) = \sum_{i=1}^n [\delta_i \ln f(t_i|\theta) + (1 - \delta_i) \ln S(t_i|\theta)] \quad (3)$$

The combinatorial logic of Equation (3) strictly distinguishes known from unknown states: n is the total number of equipment; θ represents the distribution parameters to be estimated (such as the shape and scale parameters of Weibull); δ_i is the failure event indicator (if explicit physical failure occurs, $\delta_i = 1$; if only safe operating time is recorded by the end of the observation period and it is right-censored, then $\delta_i = 0$).

The additive logic of this equation (corresponding to the continuous multiplication of the original likelihood function in the logarithmic domain) lies in the isomorphic integration of two completely different physical information types: for failed individuals where $\delta_i = 1$, since we know their exact time of death, we extract their probability density value $f(t_i|\theta)$ at time t_i ; whereas for un-failed individuals where $\delta_i = 0$, although their ultimate lifespan is unknown, the physical fact is that “the equipment survived at least until t_i ,” thus we extract its survival probability $S(t_i|\theta)$ (i.e., the total probability constraint of failing at or after t_i).

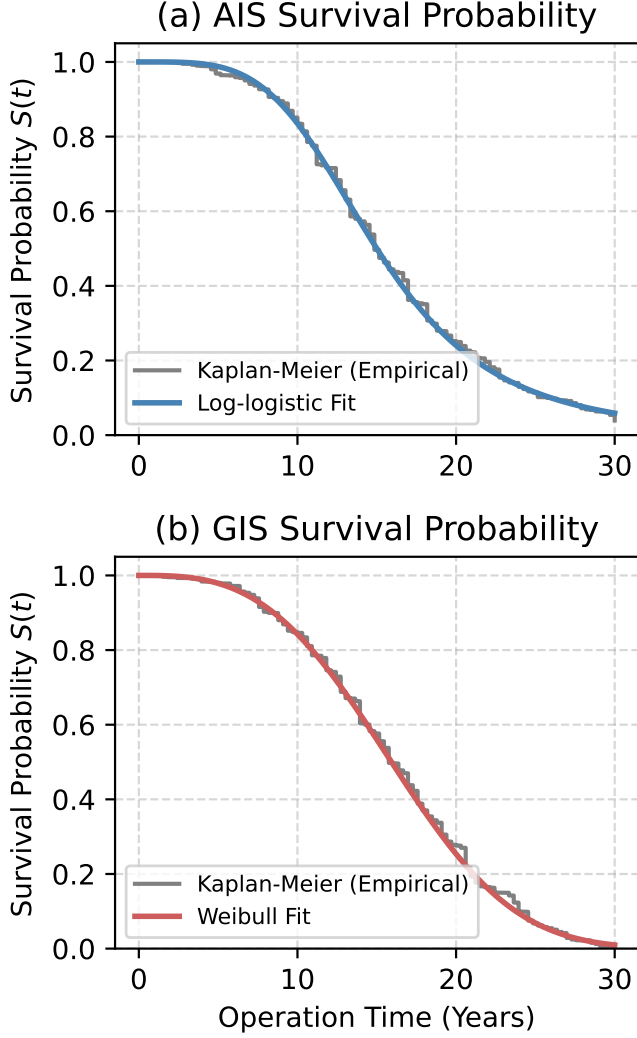


Fig. 2. Comparison chart of survival probability $S(t)$ based on Kaplan-Meier empirical estimation and parametric model fitting. (a) AIS data with Log-logistic model fitting; (b) GIS data with Weibull model fitting. The step-like broken lines represent the simulated observation means containing actual right-censored events, while the smooth curves represent the theoretical distributions given by the models. The high degree of alignment between the two confirms the physical validity of the underlying distribution parameter selections.

Through this mathematical mechanism, the system completely absorbs the true service characteristics of the asset population without tampering with any underlying labels or discarding un-failed samples, thereby providing an unbiased data foundation for subsequent panel diagnostics.

To intuitively demonstrate the statistical effect of MLE after processing censored data, the system, based on numerical simulation data constructed via Monte Carlo sampling (containing both exact failures and right-censored records), plotted a survival probability $S(t)$ step-comparison chart based on Kaplan-Meier nonparametric estimation [21], as shown in Fig. 2.

The horizontal axis of Fig. 2 represents equipment operating time, and the vertical axis represents the survival probability $S(t)$ of the equipment population, i.e., the probability that the

equipment has not failed at a specific point in time. Visually, the chart presents two different structural forms of lines: a step-like broken line with a gray shaded band, and colored smooth curves interspersed within.

The step-like broken line represents the non-parametric observation empirical mean based on simulation data. Because equipment failure or censored clock-in events occur discretely on the time axis in complex physical conditions, every time an event is recorded, the population survival probability experiences a step-like drop, forming a stepped appearance. The gray shaded area surrounding the step line represents the 95% Confidence Interval calculated based on existing observation samples. This interval reflects the statistical fluctuation range caused by limited sampling quantities, indicating that the true overall survival probability fluctuates up and down within this banded area with high confidence.

On the other hand, the colored smooth curves in Fig. (a) and Fig. (b) respectively represent the theoretical continuous survival probability distribution trajectories calculated by the system after fitting the Log-logistic model and the Weibull model using MLE. From the two subplots, it can be observed that the smooth curves representing theoretical predictions closely follow the step-like observation means in overall downward trend, and consistently remain within the statistical confidence intervals marked by the gray shading.

This visual comparison corroborates two inferences: First, it confirms that the underlying continuous probability distributions selected in this study (Log-logistic and Weibull) can objectively correspond to the actual degradation trajectories of AIS mechanical fatigue and GIS chemical aging; Second, it verifies that the MLE mathematical mechanism adopted by the system, without discarding “undamaged surviving equipment,” effectively corrects survivorship bias, providing an unbiased data base for subsequent multivariate model auditing.

B. 6-Mirror Horse-Racing Mechanism and Evaluation Matrix

The system does not presuppose any single model but constructs a “racing panel” comprising six heterogeneous survival algorithms. These six mirrors cover semi-parametric models (such as the classic Cox Proportional Hazards model [22]), fully parametric models (Weibull, Log-logistic, Lognormal, Exponential), and non-linear machine learning models (such as Random Survival Forests, RSF [23]). The general selection and classification architecture of these models are cited from [4].

To objectively evaluate the fitting quality of the six models in the panel to the data, the system introduces core evaluation metrics specifically designed for censored data. Conventional Mean Squared Error (MSE) or Accuracy often fails when facing time series with right-censored attributes. Therefore, the system adopts the Concordance Index (C-index [24]) proposed by Frank Harrell and the Brier Score [25].

Taking the C-index, which evaluates a model’s discriminative ability, as an example, its academic origins (meta-equation) can be traced back to **Kendall’s Tau rank correlation coefficient** [26] and the **Area Under the ROC Curve (AUC)** [27] in nonparametric statistics. Its core function is

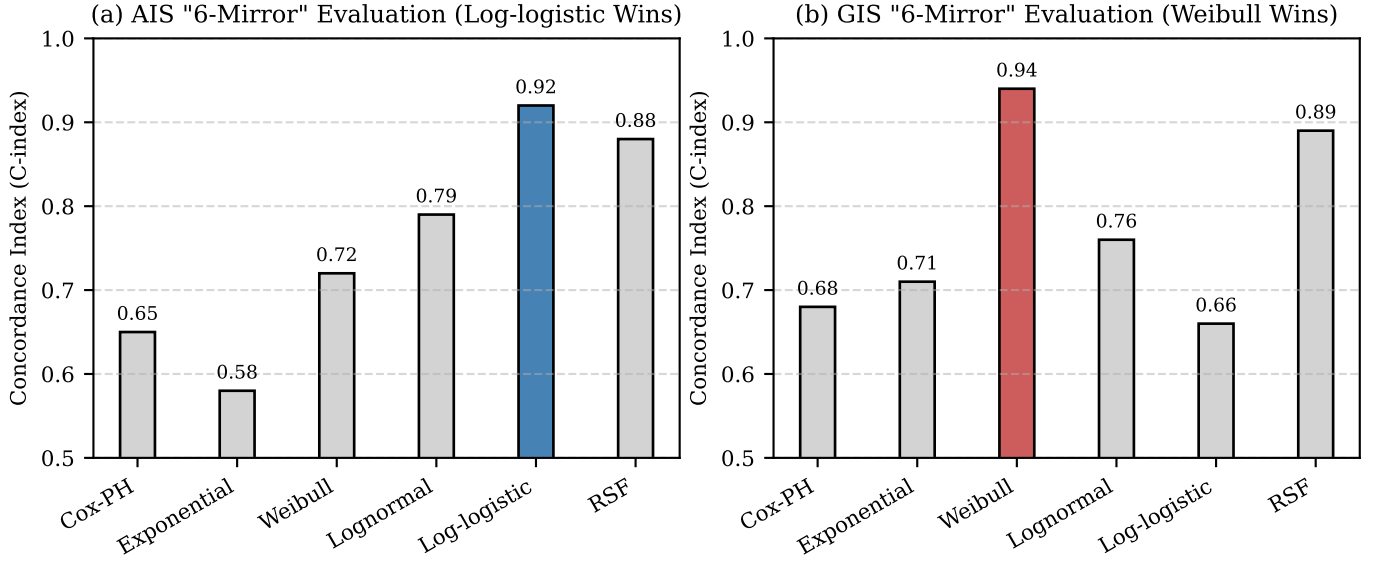


Fig. 3. C-index evaluation results of the “6-Mirror” panel diagnostics for two types of switchgear based on Monte Carlo sampling simulation. (a) AIS horse-racing results, with the Log-logistic model winning with the highest C-index; (b) GIS horse-racing results, with the Weibull model winning. All chart data originate from numerical calculus of underlying mathematical laws, not actual grid field measurements.

not to predict the exact failure days of the equipment, but to evaluate the model’s global relative ranking ability to predict “which equipment will fail first.” To accommodate censored conditions, its mathematical deconstruction is as follows:

$$C = \frac{\sum_{i,j} I(T_i > T_j) \cdot I(\eta_i < \eta_j) \cdot \delta_j}{\sum_{i,j} I(T_i > T_j) \cdot \delta_j} \quad (4)$$

The combinatorial logic of Equation (4) demonstrates a multiplicative filtering mechanism for global data Pairs. Here, $I(\cdot)$ is a boolean indicator function; T represents the actual observed survival time; the continuous variable η represents the risk prediction score output by the model; δ is the most crucial event indicator (set to 0 if the equipment is right-censored due to non-failure, and set to 1 if a clear physical failure occurs).

The core mechanism of this equation in handling right-censored data lies in multiplicative filtering: only when equipment j is an individual with a definite failure (i.e., $\delta_j = 1$) and the observed lifespan of equipment i is longer than that of j (i.e., $T_i > T_j$), is this pair of samples regarded as a “Comparable Pair” in the denominator. Under the premise of effective comparison, if the risk score predicted by the model for equipment i with a longer lifespan is indeed lower than that for equipment j with a shorter lifespan (i.e., $\eta_i < \eta_j$), then the product of all indicator functions in the numerator becomes 1, recorded as one correct prediction (Concordance).

Through this matching and division conversion of underlying data, the C-index result is strictly mapped into the $[0.5, 1.0]$ interval: 0.5 represents random blind guessing equivalent to flipping a coin, and 1.0 represents that the model accurately locks onto the objective physical sequence of asset degradation. The system relies precisely on this hard mathematical metric to objectively screen out the winner algorithm in the panel that best matches the asset’s physical fingerprints.

C. White-box Causal Auditing and Numerical Simulation Validation

By allowing the six models to compete blindly in a simulated dataset generated based on the Monte Carlo method [28] with normal noise superimposed, we can reversely corroborate the deep physical effects of the equipment using the specific statistical distribution laws of the ultimate winning model. In the numerical deduction, the competitive results are shown in Fig. 3.

This full-width twin figure displays the C-index scores of the six models when processing the two core assets (the higher the bar height, the stronger the model’s discriminative ability to predict degradation sequences). The horizontal axis lists the six mirrors participating in the race, and the vertical axis is the numerical range of the C-index $[0.5, 1.0]$. The gray bars represent the defeated conventional models, while the bars with specific values at the top and marked in special colors represent the ultimate winners.

In the AIS audit of Fig. 3(a), the Log-logistic model represented by the Steelblue bar stands out with a C-index as high as 0.92, significantly outperforming the widely used Cox model (0.65) and the non-linear RSF model (0.88). This significant relative ranking discriminative ability proves that this distribution model objectively aligns with the trend of real data. This purely data-driven blind selection result objectively corroborates mathematically the nonlinear “middle-age crisis” faced by AIS mechanical operations mentioned earlier.

Conversely, in the GIS audit of Fig. 3(b), the Weibull model represented by the Indianred bar wins with a score of 0.94. Due to the inherent unidirectional attenuation attribute of the Weibull distribution, this highly consistent ranking discriminative ability intuitively supports that GIS chemical seal aging is an irreversible, continuously deteriorating physical process. Through this blind-selection mechanism, the system

sheds its reliance on the subjective prior experience of grid O&M experts, completing the physical reverse deconstruction of the underlying degradation truths of two heterogeneous equipments relying solely on mathematical laws.

IV. DYNAMIC SOFT-THRESHOLD APPROACHING BASED ON THE P-F INTERVAL

In the conservative grid environment, waiting for an asset to drop below the functional death redline (F-point, i.e., Functional Failure, whose dead threshold, as mentioned earlier, is controlled by authoritative industrial standards like IEC and factory test baselines) to validate an algorithm is accompanied by exorbitant economic costs. This system proposes a “Data + Time” dual-insurance trigger mechanism to define the potential failure starting point (P-point), and executes closed-loop calibration and retirement economic evaluation between the P-point and the F-point.

A. 10-Decile Stepped Calibration Mechanism

In reliability engineering, the degradation trajectory of equipment from the potential failure starting point (P-point, where performance begins to clearly deviate from the healthy baseline) to the ultimate functional failure endpoint (F-point, the dead threshold redline) is full of nonlinearities and uncertainties. To map abstract physical degradation into concrete digital algorithms, the system first defines this degradation redline at the sensor level, and its evolution process is shown in Fig. 4.

To achieve high-precision lifespan tracking within this P-F interval, the system discards the static setting where a single prediction determines the lifetime outcome, and introduces a 10-Decile sequential soft-threshold calibration mechanism. It must be specifically pointed out that the dynamic tracking of individuals in this section and the macroscopic population audit results in Section III constitute a rigorous causal closed loop: the physical distribution laws and their underlying parameters blindly selected by the “6-Mirror Horse Race” on the large-scale sample in the previous chapter (i.e., the Log-logistic parameters for AIS and Weibull parameters for GIS α, β) will be directly used as the **[Prior Distribution]** for the Bayesian updating equation in this section. Only on this prior base does the system truly initiate individualized degradation correction.

Specifically, once the equipment’s online monitored physical fingerprint (such as circuit breaker spring operation delay time) crosses the P-point in Fig. 4 (i.e., the top orange dashed line intersection) triggering an initial alarm, the system instantly divides the physical residual space between the equipment’s current degradation level and the ultimate functional death redline (F-point, i.e., the bottom thick red solid line, such as the limit lock value specified by IEC standards) into 10 degradation steps (Deciles). As visually mapped in the figure, the gray scatter points represent real sensor feedback data filled with operating noise, and the solid dark-red line is the main degradation trajectory extracted by the algorithm after peeling off the noise; the 9 vertical clock-in dividing lines that divide the spatial space consist of the first 8 dotted blue

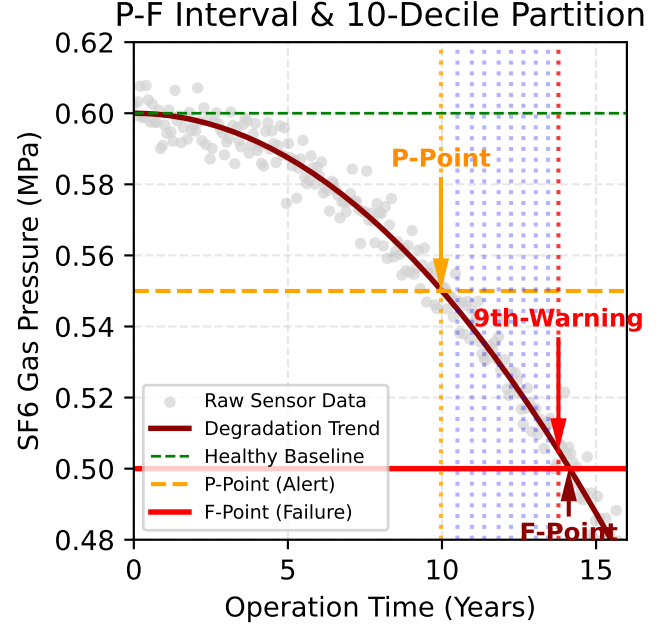


Fig. 4. P-F physical degradation trajectory and 10-Decile stepped division based on sensor data. Gray scatter points represent original sensor observations superimposed with operating condition noise, and the solid dark-red line is the main degradation trajectory after stripping noise. From left to right, the image is controlled by three core trigger points: the intersection where the solid dark-red line cuts down across the top horizontal dashed warning line is the [P-point (warning start)]; the intersection cutting right across the 9th vertical dotted red line is the [9th Decile (mandatory scheduling alarm point)]; and finally plunging into the intersection of the bottom thick solid red line is the [F-point (functional death redline)]. Between the P-point and the 9th decile, 8 vertical dotted blue lines are evenly distributed, representing safety clock-in nodes used for Bayesian correction and economic optimization. The data in this figure also originates from numerical simulation deductions superimposed with Gaussian noise.

lines representing economic optimization observation nodes, and the final 1 dotted red line (9th-Warning) representing a mandatory scheduling alarm.

Subsequently, when the equipment’s dark-red degradation trajectory sequentially crosses the actual moments of these 9 clock-in lines downwards, the system executes a closed-loop calibration every single time: First, the model calculates the expected time span for the equipment to reach the next clock-in line based on the survival curve at the current truncation time; Second, when sensor data confirms that the equipment has actually arrived there, the system measures the residual between the “predicted time” and the “actual arrival time” as the latest physical observation fact, and inputs it into the Sequential Bayesian Updating [29] equation. Its meta-equation is deconstructed as follows:

$$P(\theta|\mathcal{D}_{1:k}) = \frac{P(\mathcal{D}_k|\theta) \cdot P(\theta|\mathcal{D}_{1:k-1})}{\int P(\mathcal{D}_k|\theta) \cdot P(\theta|\mathcal{D}_{1:k-1})d\theta} \quad (5)$$

Equation (5) elucidates the multiplicative logic of knowledge accumulation: θ is the core degradation parameter of the survival model; $\mathcal{D}_k$ is the true sensor observation time when the equipment actually crosses the k -th step. In the numerator, $P(\theta|\mathcal{D}_{1:k-1})$ represents the **prior knowledge** accumulated by the system upon crossing the previous step; $P(\mathcal{D}_k|\theta)$ is the

Likelihood of the newly observed data; the denominator is the marginal probability density, used for probability normalization under continuous distribution.

Through the “prediction-verification-update” sequence iteration constructed by Equation (5), the algorithm is able to continuously acquire 9 physical correction opportunities during the entire P-F phase as the equipment slides toward failure. Every time the equipment actually crosses a step, the product operation of Bayes’ theorem forces the correction of the drift in parameter θ . As the step k continuously advances backwards, the exponential accumulation of prior information causes the posterior Probability Density Function (PDF) regarding Remaining Useful Life (RUL) to narrow sharply on the time axis, and the Error Funnel shows a significant convergence trend. This high-frequency dynamic soft-threshold approaching provides highly reliable inputs with low variance for subsequent economic equations on a mathematical basis.

B. Optimal Stopping Theory and Deconstruction of the Age Replacement Model

1) *Dynamic Economic Boundary Evaluation (Deciles 1-8 Phase)*: At every degradation node crossing steps 1 through 8, the system acquires a low-variance posterior residual life distribution under the current node via Bayesian updating. Based on this distribution, the system invokes the classic **Age Replacement Model** (see the pioneering work of Barlow et al. [30]) within the framework of Optimal Stopping Theory [12] to recalculate economic extremes. The meta-equation for this model is deconstructed as follows:

$$C(\tau) = \frac{C_p \cdot S(\tau) + C_u \cdot (1 - S(\tau))}{\int_0^\tau S(t)dt} \quad (6)$$

The purpose of Equation (6) is to calculate the expected total cost per unit time $C(\tau)$ if the asset is replaced at service time τ .

In terms of combinatorial logic, the numerator represents the weighted sum of expected expenditure costs: C_p is the fixed cost of planned outage preventive replacement; C_u is the punitive cost of unplanned outages and equipment damage caused by sudden failures (in actual conditions $C_u \gg C_p$). $S(\tau)$ is the survival probability output by the model in real-time. The numerator applies the additive logic of mutually exclusive events: if the equipment survives (probability $S(\tau)$), only the fixed cost C_p needs to be paid; if the equipment has failed (probability $1 - S(\tau)$), the punitive cost C_u must be paid. The denominator $\int_0^\tau S(t)dt$ is the expected failure-free operating time of the equipment within time τ . By dividing the total cost by the expected operating time, the equation converts assets with different lifespans into a unified time-dimensional economic benchmark. From the perspective of industrial commercial gaming, this equation reveals a highly tense trade-off process: if replaced too early (τ is extremely small), although failure risks are completely avoided, frequent C_p expenditures will cause the average daily holding cost to surge; if asset residual value is overly extracted (τ is extremely large), although procurement costs are diluted, the sudden surge in equipment sudden death rate $1 - S(\tau)$ will cause the

mathematical expectation of facing a massive penalty C_u to explode exponentially. At every safety node crossing the first 8 deciles, the algorithm accurately searches for the optimal stopping point τ^* that minimizes $C(\tau)$ by taking the derivative of Equation (6). Once the current physical time approaches this economic extreme point, the system generates an “Optimal Economic Outage Maintenance Recommendation Report” and submits it to the dispatch center, to be manually confirmed by the owner combined with actual grid conditions.

2) *Highly Reliable Safety Fusing Boundary (9th Decile Warning and Mandatory Scheduling)*: The fundamental goal of predictive maintenance is value protection, not destructive physical experiments; the system must resolutely prevent the equipment from truly running to the F-point (i.e., the 10th decile death redline). When the equipment’s degradation indicator crosses the 9th decile (entering the final 10% of physical residual space), regardless of whether the economic extreme point calculated by Equation (6) has been reached, the system will instantly trigger a physical fusing alarm. At this time, the algorithm no longer waits for an economic crossover point, but immediately generates a “High-Risk Outage Maintenance Report,” and submits it to the owner along with the system-predicted deadline for reaching the 10th decile. The specific power outage isolation and switching operations are arranged by the owner according to grid operating conditions, but the only bottom line is: the owner must complete the power outage replacement of the equipment ahead of the system-predicted arrival of the 10th decile (F-point). This dual-track mechanism, where “the first 8 deciles rely on economic optimization to extract residual value, and the 9th decile is handed over to human decision-making with a safety net before the F-point,” not only returns the ultimate discretionary power to industrial operators but also realizes the commercial value mining of the asset’s service cycle while avoiding unplanned outage losses.

C. Numerical Validation Map of Economic Optimal Stopping Point and Error Convergence

To intuitively display the linkage process of the aforementioned 10-Decile sequential Bayesian calibration and age replacement model in the true physical domain, this system conducted numerical deductions based on the aforementioned distribution laws and equations, and the complete mapping results are shown in Fig. 5.

In Fig. 5, the left subplot (a) exhaustively depicts the Bayesian convergence process of Equation (5). It should be particularly pointed out that this funnel does not expand from the initial point of the horizontal axis but strictly “detonates” from the instant the P-point alarm is triggered. This confirms in engineering logic the system’s “silent monitoring” mechanism during the healthy period. The error funnel formed by the light-blue shaded area demonstrates the algorithm’s convergence evolution towards uncertainty: in the early stage when the P-point is just triggered, due to the short historical physical path available for reliance, the funnel presents a divergent bell-mouth state; but as the equipment continuously advances on the horizontal axis and sequentially triggers the red scatter points (i.e., true observation data of reaching each step), the

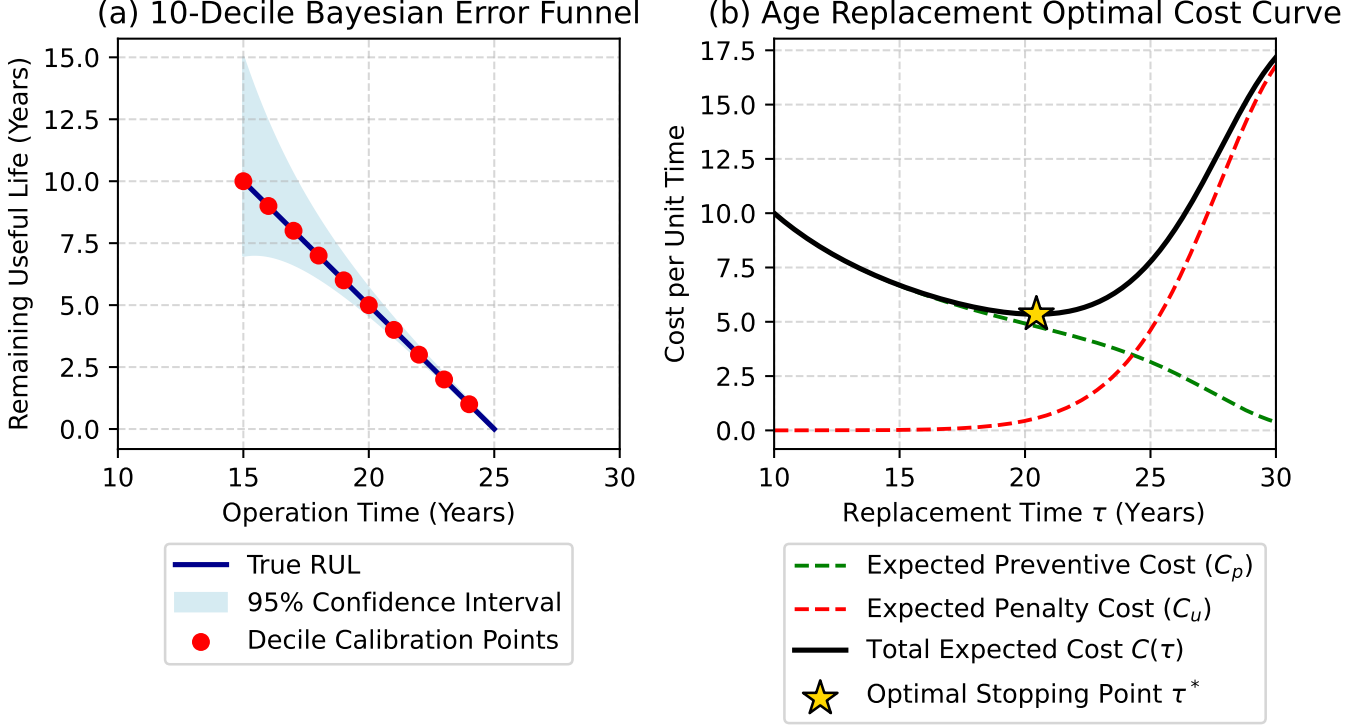


Fig. 5. Twin figures of RUL error convergence funnel and optimal stopping theory economic optimization based on 10-Decile Bayesian updating. (a) Predictive error funnel convergence characteristics: horizontal axis is equipment operating time, vertical axis is Remaining Useful Life (RUL). The solid dark-blue line represents the true ultimate lifespan limit point, the light-blue gradient area represents the 95% confidence interval deduced by Bayesian posterior probability, and the red scatter points are the true observation clock-in points triggered when the equipment crosses each step; (b) Age Replacement Model $C(\tau)$ cost extreme curve: horizontal axis is subsequent intervention time, vertical axis is expected total cost. The green dashed line represents fixed preventive cost C_p , the red dashed line represents expected failure penalty cost climbing gradually with degradation, the dark solid line represents comprehensive expected total cost per unit time, and the pentagram marks the optimal theoretical extreme point τ^* locked and triggered for operation via differential calculus. All data are generated by numerical simulations obeying the aforementioned mathematical mechanisms.

addition of likelihood in Equation (5) forces the variance of the posterior probability distribution to shrink sharply. This data morphology directly proves that multiple closed-loop clock-in calibrations can effectively eliminate long-term prediction uncertainty blind spots.

The right subplot (b) is the mathematical manifestation of the Equation (6) age replacement model. The figure clearly decomposes the gaming relationship of costs through a curve intersection structure: as the equipment operates with disease, the average daily depreciation curve representing the fixed preventive cost C_p (Green dashed line) amortized over time continues to decline; however, due to continuously approaching the F-point, the explosive failure penalty risk represented by the red dashed line starts to soar exponentially. The dark solid line represents the $C(\tau)$ U-shaped trajectory after the addition of the two costs. By taking the derivative of the underlying formula of this curve, the system accurately locks onto the pentagram-shaped extreme point (τ^*) located at the bottom of the valley. In the numerical simulation, this theoretical optimal intervention time falls precisely within the safe life interval between the P-point and the F-point, thereby providing grid dispatch departments with an optimal outage operation schedule backed by rigorous statistical conversion and clear commercial ROI.

V. CONCLUSION

By integrating survival algorithms with microscopic causal auditing, this study established an asymmetric predictive maintenance commercial boundary based on “outage isolation costs,” successfully upgrading traditional calendar-based population life management to precise individualized dynamic asset management. Through the “Tale of Two Cities” framework, data and mathematics were used to reversely prove the non-monotonic mechanical fatigue of AIS and the monotonic chemical aging characteristics of GIS, ultimately providing grid asset managers with a highly engineering-feasible precise life-extension and dynamic retirement economic decision redline.

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In the process of exploring underlying survival algorithms for predictive maintenance of distribution grid assets, this study was profoundly inspired by the review literature [4] published by M. R. Shadi et al. This outstanding review article provided a comprehensive, systematic, and in-depth retrospective on multi-dimensional survival models in sensor-enabled smart energy networks. Through detailed deconstruction and comparison of fully parametric models, semi-parametric models, and machine learning/deep learning-based

survival analysis architectures, the original author team provided academia with a valuable algorithmic panorama and theoretical summary.

It is precisely thanks to the high-level perspective and meticulous classification guidance of this literature that this study was able to smoothly establish the “6-Mirror Panel Auditing” mechanism of the *Bianque System* and accurately complete the selection of the underlying core algorithm library. Herein, we express our most sincere respect and gratitude to the original author team represented by M. R. Shadi. Their high-level research achievements laid a solid theoretical cornerstone for this paper.

REFERENCES

- [1] K. H. M. Azmi, N. A. M. Radzi, N. A. Azhar, et al., “Active electric distribution network: applications, challenges, and opportunities,” *IEEE Access*, vol. 10, pp. 134655-134689, 2022. [Online]. Available: <https://doi.org/10.1109/ACCESS.2022.3229328>
- [2] H. Mirshekali, L. K. Mortensen, and H. R. Shaker, “Reliability-aware multi-objective approach for predictive asset management: A Danish distribution grid case study,” *Applied Energy*, vol. 358, p. 122556, 2024. [Online]. Available: <https://doi.org/10.1016/j.apenergy.2023.122556>
- [3] M. Foata, “Economics of transformer and reactor life management,” in *Transformer and Reactor Life Management*, Cham: Springer Nature Switzerland, 2024, pp. 79-114. [Online]. Available: https://doi.org/10.1007/978-3-031-77219-1_4
- [4] M. R. Shadi, H. Mirshekali, M. Tahavori, and H. R. Shaker, “Survival Models for Predictive Maintenance and Remaining Useful Life in Sensor-Enabled Smart Energy Networks: A Review,” *Sensors*, vol. 26, no. 6, p. 1915, 2026. [Online]. Available: <https://doi.org/10.3390/s26061915>
- [5] M. S. Alvarez-Alvarado, D. L. Donaldson, A. A. Recalde, et al., “Power system reliability and maintenance evolution: A critical review and future perspectives,” *IEEE Access*, vol. 10, pp. 51922-51950, 2022. [Online]. Available: <https://doi.org/10.1109/ACCESS.2022.3172697>
- [6] Y. Zeng, “From Causal Auditing to Value Maximization: Individualized Management Guidelines for Offshore Wind Assets (v1.0.0),” *Zenodo*, 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.20391481>
- [7] Y. Zeng, “Mitigating Large Language Model Hallucinations in Industrial Environments via Asymmetric Deterministic Override Architecture,” *Zenodo*, 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.20539607>
- [8] IEC 62271-100, “High-voltage switchgear and controlgear - Part 100: Alternating-current circuit-breakers,” *International Electrotechnical Commission*, Geneva, Switzerland. [Online]. Available: <https://webstore.iec.ch/en/publication/62785>
- [9] Y. Tan, E. Hu, Y. Liu, et al., “Review of digital vibration signal analysis techniques for fault diagnosis of high-voltage circuit breakers,” *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 31, no. 1, pp. 404-418, 2023. [Online]. Available: <https://doi.org/10.1109/TDEI.2023.3339114>
- [10] IEC 62271-203, “High-voltage switchgear and controlgear - Part 203: Gas-insulated metal-enclosed switchgear for rated voltages above 52 kV,” *International Electrotechnical Commission*, Geneva, Switzerland. [Online]. Available: <https://webstore.iec.ch/en/publication/65853>
- [11] M. M. Alomari, A. Bani-Mustafa, A. W. B. Dirawieh, et al., “SF₆ leakage risk in gas-insulated switchgear under the interaction of ageing and environmental stress,” *Results in Engineering*, p. 109929, 2026. [Online]. Available: <https://doi.org/10.1016/j.rineng.2026.109929>
- [12] Y. Yatsenko and N. Hritonenko, “Analytics of machine replacement decisions: economic life vs real options,” *Management Decision*, vol. 60, no. 2, pp. 471-487, 2022. [Online]. Available: <https://doi.org/10.1108/MD-12-2019-1704>
- [13] Y. Zeng, “From Causal Auditing to Value Maximization for Core Distribution Assets—Part I: Underground Cable Management (v1.0.0),” *Zenodo*, 2026. [Online]. Available: <https://doi.org/10.5281/zenodo.20676445>
- [14] F. S. Nowlan and H. F. Heap, “Reliability-centered maintenance,” United Airlines and U.S. Department of Defense, Washington, DC, USA, Rep. AD-A066579, 1978. [Online]. Available: <https://apps.dtic.mil/sti/html/tr/ADA066579/>
- [15] M. Rausand and A. Hoyland, *System Reliability Theory: Models, Statistical Methods, and Applications*. Hoboken, NJ, USA: John Wiley & Sons, 2003.
- [16] D. Collett, *Modelling Survival Data in Medical Research*. Boca Raton, FL, USA: Chapman and Hall/CRC, 2023. [Online]. Available: <https://doi.org/10.1201/9781003282525>
- [17] L. Dou, S. Liu, X. Pu, et al., “Research on online detection method of high voltage circuit breaker closing spring fatigue based on energy storage action model,” *Electric Power Systems Research*, vol. 248, p. 111985, 2025. [Online]. Available: <https://doi.org/10.1016/j.epsr.2025.111985>
- [18] M. J. Crowther, P. Royston, and M. Clements, “A flexible parametric accelerated failure time model and the extension to time-dependent acceleration factors,” *Biostatistics*, vol. 24, no. 3, pp. 811-831, 2023. [Online]. Available: <https://doi.org/10.1093/biostatistics/kxac009>
- [19] C. She, F. Zeng, H. Li, et al., “Diffusion and absorption behaviors of small gas molecules in EPDM and NBR sealing materials for C₅F₁₀O insulated switchgears,” *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 31, no. 2, pp. 722-730, 2023. [Online]. Available: <https://doi.org/10.1109/TDEI.2023.3319593>
- [20] M. Maswadah, “Improved maximum likelihood estimation of the shape-scale family based on the generalized progressive hybrid censoring scheme,” *Journal of Applied Statistics*, vol. 49, no. 11, pp. 2825-2844, 2022. [Online]. Available: <https://doi.org/10.1080/02664763.2021.1924638>
- [21] E. L. Kaplan and P. Meier, “Nonparametric estimation from incomplete observations,” *Journal of the American Statistical Association*, vol. 53, no. 282, pp. 457-481, 1958. [Online]. Available: <https://doi.org/10.1080/01621459.1958.10501452>
- [22] D. R. Cox, “Regression models and life-tables,” *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 34, no. 2, pp. 187-202, 1972. [Online]. Available: <https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>
- [23] H. Ishwaran, U. B. Kogalur, E. H. Blackstone, and M. S. Lauer, “Random survival forests,” *The Annals of Applied Statistics*, vol. 2, no. 3, pp. 841-860, 2008. [Online]. Available: <https://doi.org/10.1214/08-AOAS169>
- [24] F. E. Harrell, R. M. Califf, D. B. Pryor, K. L. Lee, and R. A. Rosati, “Evaluating the yield of medical tests,” *JAMA*, vol. 247, no. 18, pp. 2543-2546, 1982. [Online]. Available: <https://doi.org/10.1001/jama.1982.03320430047030>
- [25] G. W. Brier, “Verification of forecasts expressed in terms of probability,” *Monthly Weather Review*, vol. 78, no. 1, pp. 1-3, 1950. [Online]. Available: [https://doi.org/10.1175/1520-0493\(1950\)078%3C0001:VOFEIT%3E2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078%3C0001:VOFEIT%3E2.0.CO;2)
- [26] M. G. Kendall, “A new measure of rank correlation,” *Biometrika*, vol. 30, no. 1/2, pp. 81-93, 1938. [Online]. Available: <https://doi.org/10.1093/biomet/30.1-2.81>
- [27] A. P. Bradley, “The use of the area under the ROC curve in the evaluation of machine learning algorithms,” *Pattern Recognition*, vol. 30, no. 7, pp. 1145-1159, 1997. [Online]. Available: [https://doi.org/10.1016/S0031-3203\(96\)00142-2](https://doi.org/10.1016/S0031-3203(96)00142-2)
- [28] E. Zio, “Monte Carlo simulation: The method,” in *The Monte Carlo Simulation Method for System Reliability and Risk Analysis*, London, U.K.: Springer, 2013, pp. 19-58. [Online]. Available: https://doi.org/10.1007/978-1-4471-4588-2_3
- [29] N. Chen and K.-L. Tsui, “Condition monitoring and remaining useful life prediction using degradation signals: Revisited,” *IIE Transactions*, vol. 45, no. 9, pp. 939-952, 2013. [Online]. Available: <https://doi.org/10.1080/0740817X.2012.706376>
- [30] R. Barlow and L. Hunter, “Optimum preventive maintenance policies,” *Operations Research*, vol. 8, no. 1, pp. 90-100, 1960. [Online]. Available: <https://doi.org/10.1287/opre.8.1.90>